PoML Exercise session 1

Decision trees and K-nearest-neighbors

Exercise types

- Drawing
  - Exercise 1: Drawing trees for binary variables
  - Exercise 2: Drawing trees for numeric variables
  - Exercise 7: Drawing Voronoi maps for kNN decision boundaries
- Calculation
  - Exercise 3: Finding the best variable to split on for a decision tree
  - Exercise 4: Applying a tree-generating algorithm
  - Exercise 6: Applying k-NN with non-standard distance metrics
- Intuition and reasoning:
  - Exercise 5: going beyond binary prediction in decision trees
  - Exercise 8: kNN and curse of dimensionality
- Coding
  - Exercise 9: Applying k-NN on a real dataset

Exercise 1.3.a

- Entropy is defined as $CE(D) = -\sum_{k=1}^{K} \frac{|D_k|}{|D|} \log_2 \left(\frac{|D_k|}{|D|} \right)$ with D the the data points, and D_k the data points that belong to class k.
- For label y_1, 3 points belong to class $\oplus$ and 3 to $\ominus$
- So $D_{\oplus} = \{1,2,4\}$ and $D_{\ominus} = \{3,5,6\}$
- So

$$CE(D) = -\left(\frac{|D_{\oplus}|}{|D|} \log_2 \left(\frac{|D_{\oplus}|}{|D|} \right) + \frac{|D_{\ominus}|}{|D|} \log_2 \left(\frac{|D_{\ominus}|}{|D|} \right) \right) = -\left(\frac{3}{6} \log_2 \left(\frac{3}{6} \right) + \frac{3}{6} \log_2 \left(\frac{3}{6} \right) \right) = -\left(-\frac{1}{2} - \frac{1}{2} \right) = 1$$

- Entropy measures variation, chaos, overall difference, etc.

Exercise 1.3.b

- Information gain intuitively:
  - Calculate E(current node), i.e. chaos of current situation
  - Split on branch: calculate E(current node -> go left) and E(current node-> go right)
  - Scale these values proportional to their size: E(left branch) = (|left samples|/|samples in current node|)* E(current node -> go left)
  - E(current node) - E(left branch) - E(right branch)> 0? This reduces chaos, good split!
  - E(current node) - E(left branch) - E(right branch) < 0? This increases the chaos, bad split!

Exercise 1.3.b (continued)

- Information gain is defined as $IG(D,t) = CE(D) - \sum_{o=1}^{O} \frac{|D_o|}{|D|} CE\left(D_o\right)$ with t a test (e.g. splitting on a_2)
- Split on variable a_2: instances 3,4 go left, 1,2,5,6 go right
  - So $D_{a_2=F} = \{3,4\}$ and $D_{a_2=T} = \{1,2,5,6\}$
- $CE\left(D_{a_2=F}\right) = -\left(\frac{|D_{\oplus,a_2=F}|}{|D_{a_2=F}|}\log_2\left(\frac{|D_{\oplus,a_2=F}|}{|D_{a_2=F}|}\right) + \frac{|D_{\ominus,a_2=F}|}{|D_{a_2=F}|}\log_2\left(\frac{|D_{\ominus_{a_2=F}}|}{|D_{a_2=F}|}\right)\right)$
- In $D_{a_2=F}$, one sample has class $\oplus$ and one has $\ominus$. So the probabilities are equal again!
  - $CE\left(D_{a_2=F}\right) = 1$
- In $D_{a_2=T}$, there are 4 samples, but again equal probabilities. so $CE\left(D_{a_2=T}\right) = 1$
- Therefore, $IG\left(D,a_2\right) = 1 - \left(\frac{2}{6}\cdot 1 + \frac{4}{6}\cdot 1\right) = 0$

Exercise 1.3b (continued)

- Step 1: calculate $IG\left(D, a_1\right)$ when I tell you
  - $C\left(D_{a_1=F}\right) = C\left(D_{a_1=T}\right) \approx 0.9183$
  - $IG\left(D, a_1\right) \approx 0.0817.$
- Step 2: Now show that $C\left(D_{a_1=F}\right) = C\left(D_{a_1=T}\right) \approx 0.9183.$
- Continue with 1.3.c: variance = entropy, and variance reduction = information gain

On variance formulas

- The slides toledo contain the following formula: $Var(D) = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$
- But at intro to statistics you may have learned this instead: $Var(D) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2$
- The first is also known as the **sample variance**, it assumes you might be working with small datasets. It is more robust to outliers.
- The second is known as the **population variance**. If your sample is large enough, the difference between sample and population will be small.
- Both are correct. But if you use one for an exercise, be consistent with which one you use.